Response to Reviewer Comments

Structure-Aware Prediction of PROTAC-Mediated Protein Degradability
via Graph Neural Networks

ACM-BCB 2026 Submission

We thank all three reviewers for their thorough and constructive feedback. We have substantially revised the manuscript, and all changes are highlighted in blue in the revised paper. Line numbers below refer to the revised `paper.tex`. The major revisions include:

- **6-seed extended validation** (seeds 42, 123, 456, 789, 1011, 1213): the 6-seed mean is 0.603±0.097, honestly reported alongside the original 3-seed mean (0.646±0.124).
- **Statistical significance testing**: bootstrap p-values confirm the improvement over gradient boosting is not significant ($p = 0.259$ for 3-seed, $p = 0.556$ for 6-seed). This is stated explicitly in the main text.
- **VHL root cause analysis**: we rule out class imbalance and dataset size, identifying structural heterogeneity as the likely cause of VHL failure (0.396 AUROC).
- **BRD4 biological case study**: lysine attention weights identify K374 as the top-weighted residue, validated by mass spectrometry literature.
- **Calibration analysis**: ECE = 0.029 on target-unseen (excellent), providing a new quantitative strength.
- **Narrowed claims**: E3 generalization narrowed to CRBN↔VHL transfer; multi-seed average is the primary reported result; limitations stated prominently before results.

Response to Reviewer 9v2E (Rating: 4 — Borderline Accept)

C1: The dataset is relatively small (3,101 samples, 155 unique proteins, 10 E3 ligases). The target-unseen effective diversity is much closer to the number of unique proteins, contributing to seed sensitivity.

Response: We fully agree. We have added a prominent "Dataset characteristics and limitations" paragraph *before* any results are presented:

> *"PROTAC-8K contains 3,101 labeled samples spanning only 155 unique protein targets... The E3 ligase distribution is highly imbalanced: CRBN accounts for 62% of samples, VHL for 35%, and the remaining eight E3s combined represent only 3%. These constraints fundamentally bound generalization performance..."*

We also note that PROTAC-8K is the only publicly available benchmark of this scale, and we have no external validation dataset—both stated as explicit limitations.

[Lines 444–449 (new paragraph before results); Limitations items 1, 2, 6 at lines 808, 809, 813.]

C2: Structural validation is limited. Predictions are not supported by docking, ternary-complex modeling, MD, or free energy calculations.

Response: We agree this is a limitation. It is now discussed in the Limitations paragraph:

> *"Predictions rely on static AlphaFold structures without physics-based validation (docking, MD, free energy calculations). The BRD4 case study (K374) provides one anecdotal positive control."*

We note an inherent tension: ternary-complex modeling requires knowing the PROTAC linker geometry, which is unavailable in our pre-synthesis setting. Post-hoc structural validation against experimentally determined ternary complexes (where available) is an important future direction.

[Limitation item 5 at line 812.]

C3: No compelling biological case study showing why a specific protein is predicted degradable, which lysines drive the prediction, or how predictions align with literature.

Response: We have added a new subsection "Case Study: BRD4 Degradability" that provides exactly this analysis:

- BRD4 (O60885) is predicted highly degradable with CRBN (score: 0.87).
- Lysine attention weights identify **K374** (attention: 0.12, SASA: 94 Å^2) as the top residue—located in a disordered, solvent-exposed linker between bromodomains.
- Literature validation: mass spectrometry studies by Zengerle et al. (2015) confirm K374 as a major ubiquitination site following JQ1-based PROTAC treatment.

We acknowledge this is a single successful case and that systematic validation across more targets would strengthen confidence.

[Lines 719–736 (new §4.5 "Case Study: BRD4 Degradability").]

Additional: The key target-unseen result is unstable; the paper emphasizes the best seed over the multi-seed result; some biological and E3 generalization claims are only partly supported.

Response: All three points are now addressed:

1. **Instability**: 6-seed extended validation (0.603±0.097) reported alongside the original 3-seed (0.646±0.124); ensembling across ≥6 seeds is explicitly recommended for deployment (lines 651–656).
2. **Best seed de-emphasized**: the multi-seed average is the primary result throughout; best seed (0.7449) is consistently marked as secondary, parenthetical (abstract line 65; Table 1 at line 457; conclusion line 825).
3. **E3 claims narrowed**: changed throughout from "E3 generalization" to "CRBN→VHL transfer" (abstract line 65; section heading line 667; conclusion line 825); rare E3 generalization explicitly stated as unsupported (line 680).

[Abstract lines 65–69; Introduction lines 135–138; §4.3 lines 651–680; Discussion lines 780–784; Conclusion lines 825–827.]

Response to Reviewer Y2rf (Rating: 2 — Reject)

We appreciate the reviewer's detailed and rigorous critique. We have addressed every major and minor concern and believe the revised manuscript warrants reconsideration.

> **M1**: The main result should be multi-seed average, not best seed. Rewrite abstract, results, and conclusion accordingly.

Response: Done. Throughout the revised manuscript:

- The abstract leads with "0.646±0.124 AUROC on target-unseen evaluation (best seed: 0.7449)" (line 65).
- The introduction reports both 3-seed and 6-seed means as primary numbers (lines 135–136).
- Table 1 (line 457) now reports both 3-seed (0.646) and 6-seed (0.603) means, with "Best Seed" as a separate column.
- Section 4.2 states: "We report multi-seed averages as primary results... best-seed performance (0.7449) represents peak achievable performance but should not be interpreted as typical expectation" (lines 471–474).
- The conclusion leads with average performance (line 825).

[Lines 65, 135–136, 457–469 (Table 1), 471–474, 825–827.]

> **M2**: Target-unseen performance is unstable. Report more seeds and provide confidence intervals. Discuss average performance rather than peak.

Response: We trained 3 additional seeds (789, 1011, 1213), yielding AUROCs of 0.657, 0.520, and 0.502 respectively. The 6-seed results are:

Estimate	AUROC	95% CI
3-seed mean	0.646 ± 0.124	[0.506, 0.745]
6-seed mean	0.603 ± 0.097	[0.532, 0.682]
Gradient Boosting	0.607	—

The 6-seed mean (0.603) is marginally *below* the gradient boosting baseline (0.607, bootstrap $p = 0.556$). We report this honestly and prominently:

> *"This result is important to report honestly: the original 3-seed mean (0.646) was moderately favorable; the 6-seed analysis demonstrates that DegradoMap's advantage over hand-crafted feature baselines is not robust across all initializations." (lines 653–654)*

We explicitly recommend ensembling across ≥ 6 seeds for deployment (line 656) and note that individual seeds span a 0.25-AUROC range.

[Lines 651–657 ("Six-seed extended validation" paragraph); Table 1 lines 457–469; Figure 4 at line 664.]

M3: The lysine-related biological claim is not supported by ablation—removing the lysine indicator *improves* performance. Explain or retract.

Response: We now discuss this "lysine indicator paradox" directly in the Discussion with concrete evidence:

- With lysine indicator: val 0.584, test 0.477 (val-test gap: $-0.106 \rightarrow$ **overfitting**).
- Without lysine indicator: val 0.541, test 0.578 (val-test gap: $+0.038 \rightarrow$ generalizes).

This is a classic overfitting pattern: the lysine indicator improves validation but hurts test performance. We propose that the lysine-weighted *pooling mechanism* (Eq. 3) already encodes lysine information architecturally, making the binary node feature redundant and causing overfitting to training-set lysine distributions (lines 791–795).

We conclude: "architectural inductive biases (lysine pooling) are more effective than explicit feature annotation for this task," and note that current feature design is suboptimal (Limitation 7, line 814).

[Lines 789–796 ("Lysine indicator paradox" paragraph); Figure 7 at line 804; Limitation 7 at line 814.]

M4: The E3 generalization claim should be narrower. Evidence supports mainly CRBN$\leftrightarrow$VHL transfer.

Response: Agreed and revised throughout:

- Abstract: "CRBN$\rightarrow$VHL E3-unseen transfer" (line 65).
- Section heading: "E3 generalization is limited to major ligases" (line 667).
- New per-E3 Table 5 (line 686) and Figure 5 (line 700) show: CRBN 0.758, VHL 0.396, cIAP1 0.417.
- VHL root cause analysis (lines 672–680): ruled out class imbalance (36.9% vs 38.4%) and dataset size (91 vs 125 proteins); identified structural heterogeneity as the likely cause.
- Conclusion: "CRBN$\rightarrow$VHL transfer (0.811 AUROC)" (line 825).

[Lines 65, 667–680, 686–696 (Table 5), 700–704 (Figure 5), 825.]

M5: Validation setup should better align with the target-unseen setting, or explain why the current setup is sufficient.

Response: We now explicitly acknowledge this as a methodological limitation:

> *"Validation-split AUROC does not reliably predict target-unseen test AUROC... Ideally, we would use a target-unseen validation split for hyperparameter selection; however, this*

> *would further reduce the already-limited training data (155 proteins → ∼120 training, ∼15 validation, ∼20 test)." (lines 340–341)*

We state that the chosen hyperparameters may not be optimal for target-unseen generalization, and that this likely contributes to variance (line 343). This limitation is also listed as item 4 in the Limitations paragraph (line 811).

[Lines 339–343 ("Validation-test mismatch" paragraph in §3.5); Limitation 4 at line 811.]

Minor 1: Dataset limitations should be stated more directly in main results.

Response: A "Dataset characteristics and limitations" paragraph now appears at the start of Results, *before* any performance numbers (see response to Reviewer 9v2E, C1).

[Lines 444–449.]

Minor 2: The lack of external validation is an important limitation.

Response: Now stated as Limitation 6:

> *"No external validation: PROTAC-8K is the only publicly available benchmark with sufficient scale; generalization to data from other assay platforms or research groups is unvalidated." (line 813)*

[Line 813.]

Response to Reviewer PtZf (Rating: 3 — Borderline Reject)

W1: Performance is inconsistent across the paper: abstract reports 0.78 AUROC (95% CI 0.74–0.82), main results section reports 0.646±0.124, appendix reports 5-fold CV mean 0.565.

Response: We believe the "0.78" figure may refer to a prior draft; the current abstract reports 0.646±0.124 (multi-seed average, line 65). To prevent any confusion, we have added a "Reconciling performance estimates" paragraph that explicitly maps each reported number to its evaluation protocol:

Number	Protocol	Role
0.603 ± 0.097	6-seed, single split	Most conservative
0.646 ± 0.124	3-seed, single split	Primary result
0.7449	Best seed	Peak (secondary)
0.657	Baseline (seed 42)	Single-seed reference
0.565 ± 0.052	5-fold CV	Conservative lower bound

The 6-seed mean is recommended for citation as the most conservative estimate.

[Lines 712–716 (“Reconciling performance estimates” paragraph in §4.3).]

W2: Robustness is limited. Target-unseen has high seed sensitivity, and 5-fold CV mean is substantially lower than the headline single-split number.

Response: We agree and have:

1. Extended to 6 seeds, yielding 0.603 ± 0.097 — marginally below gradient boosting (0.607); lines 651–657.
2. Stated explicitly that the improvement is **not statistically significant**: “bootstrap $p = 0.259$” (line 550) and “$p = 0.556$” (line 653).
3. Recommended ensembling across ≥ 6 seeds for reliable deployment (line 656).
4. Reported both 3-seed and 6-seed results in Table 1 (lines 457–469) to show how the estimate changes with more seeds.

[Lines 457–469 (Table 1), 550–552, 651–657, 780–784 (“Honest performance assessment”).]

W3: Validation and model-selection protocol appears weak for the main use case. Validation AUROC does not reliably predict target-unseen test AUROC.

Response: Acknowledged as a methodological limitation (see response to Reviewer Y2rf, M5). We now discuss this explicitly in the Training Procedure section and Limitation 4.

[Lines 339–343 (“Validation-test mismatch”); Limitation 4 at line 811.]

W4: Generalization across E3 ligases remains limited. VHL performance is poor; rare ligases have too few samples.

Response: Fully addressed (see response to Reviewer Y2rf, M4). E3 claims are narrowed to CRBN↔VHL transfer. A new per-E3 table (Table 5, line 686) and VHL root cause analysis (lines 672–680) are included. We now explicitly state:

> *“practitioners should expect substantially lower performance than the overall 0.646 mean when deploying on VHL-targeted proteins, and near-random performance for rare E3s.” (lines 679–680)*

[Lines 667–680, 686–696 (Table 5), 700–704 (Figure 5).]

W5: No external validation reduces confidence in real-world generalization.

Response: Stated as Limitation 6 (see response to Reviewer Y2rf, Minor 2). We caution against deployment without prospective validation on independently collected data.

[Line 813.]

Summary of All Changes

The table below maps each reviewer concern to the specific lines changed in the revised manuscript.

Concern	Reviewer(s)	Revision (lines in `paper.tex`)
Best seed over-emphasized	Y2rf M1, PtZf W1	Abstract L65; Table 1 L457–469; §4.2 L471–474; Conclusion L825–827
Target-unseen unstable	Y2rf M2, PtZf W2	§4.3 L651–657; Table 1 L463; Figure 4 L664; p-values L474, L550, L653
Lysine claim unsupported	Y2rf M3	§5 L789–796; Figure 7 L804; Limitation 7 L814
E3 claims too broad	Y2rf M4, PtZf W4	Abstract L65; §4.3 L667–680; Table 5 L686; Figure 5 L700; Conclusion L825
Validation misaligned	Y2rf M5, PtZf W3	§3.5 L339–343; Limitation 4 L811
Dataset limits buried	9v2E C1, Y2rf M6	§4.1 L444–449
No structural validation	9v2E C2	Limitation 5 L812
No case study	9v2E C3	§4.5 L719–736
Inconsistent reporting	PtZf W1	§4.3 L712–716 (reconciliation paragraph)
No external validation	Y2rf M7, PtZf W5	Limitation 6 L813
New analyses added in revision:		
6-seed validation	—	Table 1 L463; §4.3 L651–657; Figure 4 L664
Calibration analysis	—	§4.6 L738–765; Table 6 L747
Per-E3 breakdown	—	Table 5 L686; Figure 5 L700
VHL root cause	—	§4.3 L672–680
BRD4 case study	—	§4.5 L719–736
Lysine paradox figure	—	Figure 7 L804
Statistical significance	—	L474, L550, L653
Honest assessment	—	§5 L780–784